Neuromuscular and Cardiovascular Adaptations in Response to High-Intensity Interval Power Training

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[1]*Department of Physical Education and Sport, Faculty of Sport Sciences, Catholic University of Murcia (UCAM), Murcia, Spain;* [2]*ImFINE Research Group, Department of Health and Human Performance, Faculty of Physical Activity and Sport Sciences-INEF, Technical University of Madrid, Madrid, Spain; and* [3]*Hospital Virgen del Valle, Complejo Hospitalario de Toledo, Toledo, Spain*

Abstract

Romero-Arenas, S, Ruiz, R, Vera-Ibáñez, A, Colomer-Poveda, D, Guadalupe-Grau, A, and Márquez, G. Neuromuscular and cardiovascular adaptations in response to high-intensity interval power training. *J Strength Cond Res* 32(1): 130–138, 2018—The aim of this study was to determine the efficacy of a high-intensity power training (HIPT) program, and to compare the effects of HIPT to traditional power training (TPT) on the aerobic and power performance. For this purpose, 29 healthy men (23.1 ± 2.7 years) were recruited and randomly distributed into 3 different groups. One group performed TPT (n = 10), the second group performed power training organized as a circuit (HIPT; n = 10), and the third group served as control group (CG; n = 9). Training consisted of weightlifting thrice per week for 6 weeks. The TPT subjects performed 3 to 5 sets of each exercises with interset rest of 90 seconds, and HIPT subjects executed the training in a short circuit (15 seconds of rest between exercises). To know the effects in aerobic performance, maximal aerobic speed (MAS) was measured. To identify the effects on power performance, subjects performed a Wingate test, a countermovement jump (CMJ) test, and a power-load curve in bench press. The main results showed that after both power training protocols, subjects increased significantly ($p \leq 0.05$) the power production during the Wingate Test, the height and power reached during the CMJ test, and the peak power produced during the power-load curve. However, only the HIPT group improved significantly MAS ($p \leq 0.05$). There were no changes in any variables in CG. Hence, our results suggest that HIPT may be as effective as TPT for improving power performance in young adults. In addition, only HIPT elicited improvements in MAS.

Key Words high-intensity interval training, aerobic performance, muscular performance

Introduction

Recent evidence suggests that high-intensity interval training (HIIT) is a time-efficient strategy to stimulate a number of skeletal muscle adaptations that are comparable with traditional endurance training (17,32). It is well documented that HIIT may induce improvements in both anaerobic and aerobic metabolism, depending on the manipulation of the intensity and duration of bouts and recovery periods (5,20).

High-intensity interval training is characterized by brief, repeated bursts of relatively intense exercise separated by periods of rest or low-intensity exercise. One of the most common models used in low-volume HIIT studies is the Wingate Test, which consists of 30 seconds of "all-out" cycling against a high resistance on a specialized cycle ergometer (18). As little as 6 sessions of this type of training over 2 weeks robustly increase skeletal muscle oxidative capacity, as reflected by the maximal activity and (or) protein content of various mitochondrial enzymes (5,16). A modified Wingate-based HIIT protocol that consisted of 4 × 10 seconds "all-out" sprints induced improvements in aerobic and anaerobic performance that were comparable with a 4 × 30 seconds protocol (20). Another study by Metcalfe et al. (28) showed that a protocol consisting of 2 × 20 seconds "all-out" sprints, included within a 10-minute bout of primarily low-intensity cycling, improved $\dot{V}O_2$ max after 6 weeks of training (18 total sessions). The effects of HIIT have been studied for decades, the results showing that this type of training induces improvements mainly in maximal aerobic power (25,29). Furthermore, HIIT seems to improve determinant variables of endurance running performance such as velocity at maximal oxygen uptake ($\dot{V}O_2$ max) (11),

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Journal of Strength and Conditioning Research

running economy (21), velocity at lactate threshold (11), and time to exhaustion (13).

In the last years, a new variation of HIIT in combination with traditional power-strength training (based on high-intensity resistance training) using basic multijoint movements has emerged and became popular in the fitness industry. This high-intensity power training (HIPT) is performed with low rest periods between exercises, or a lack of prescribed rest periods, with the aim to finish the training as fast as possible leading to increased acute physiological demands (14). In this regard, recent research has shown that HIPT is a useful tool to obtain improvements in $\dot{V}O_2$ max and body composition (40) similar to those elicited by HIIT (18,41). It is likely that this potent high-intensity stimulus could improve aerobic and anaerobic performance; however, the higher acute physiological changes associated with this type of training could impair power output during the performance of the exercise and thus limit chronic adaptations in power performance. However, to our knowledge, it has not been yet tested the effects of an HIPT program on muscle strength and power in comparison with those obtained with a traditional power training (TPT). Traditional power training aims to improve maximal power output through the execution of varied multijoint classic resistance training exercises or generic movements such as sprinting, jumping, throwing, and kicking with the aim to produce maximal velocity in the used movement (19). Traditional power training is usually performed using few repetitions with the optimal load together with high interset rest periods to reduce changes in muscle environment, because shorter interset periods have been proven to increase peripheral fatigue (27,36), which can finally blunt strength and power adaptations (15).

In light of all this, the aim of this study was to test the effects of 6 weeks of both HIPT and TPT on the aerobic, anaerobic, and strength and power performance in healthy young active men. Specifically, although we hypothesized that HIPT training would lead to significant improvements in cardiovascular fitness, we were particularly interested in determining whether strength and power changes would be comparable with those elicited by TPT.

Methods

Experimental Approach to the Problem

A quasi-experimental pre- and post-test group design using 2 training groups and a control group (CG) to examine the short-term (6 weeks) effects of 3 sessions per week when using power training on the aerobic and power performance. Before data collection, the subjects took part in a familiarization session for each test. One week after the familiarization, the dependent variables were tested, as described below. Subsequently, the subjects were matched with respect to height, weight, and pretraining 1 repetition maximum (1RM) in bench press, and then randomly allocated to either a high-intensity power training (HIPT; $n = 10$), TPT ($n = 10$), or a CG ($n = 10$). The subjects were tested by the same investigator, using the same protocol, at the same time of day at weeks 0 and 7, and in a similar ambient temperature (19–22° C). In session 1, height and power in countermovement jump (CMJ), 1RM in bench press, load-velocity curve at 30, 40, 50, 60, and 70% of 1RM during bench press movement, and maximal aerobic speed (MAS) tests were completed. In session 2 (completed 3–4 days after session 1), 1RM in bench press, load-velocity curve at 30, 40, 50, 60, 70, and 80% of 1RM during high pull movement, and Wingate tests were completed. For the completion of all experimental protocols, the subjects were instructed to fast for 3 hours and not to consume alcohol or caffeine within 12 hours. They were also asked to avoid strenuous physical activities the day before each session. During the 6-week training period, both training groups (HIPT and TPT) performed training using a Multipower machine (Technogym SpA, Cesena, Italy) and a cycle ergometer (Bike Med, Technogym SpA) in an incremental periodized program twice a week. All subjects were asked to maintain their normal daily routines and eating habits, not to take nutritional supplements that might affect lean tissue mass, and to refrain from commencing new exercise programs during the study. One subject pertaining to the CG withdrew from the study for personal reasons.

Subjects

Twenty-nine healthy men responded to an invitation to participate in the study. All of them had at least 6 months of experience in resistance training, with a minimum frequency of 2 sessions per week. Subjects' mean ± *SD* age, height, and body mass were 23.1 ± 2.7 years old, 176.4 ± 7.9 cm, and 75.5 ± 8.7 kg, respectively. The subjects were informed about the design of the study and possible risks and discomforts related to the testing and training, after which they read and signed an informed consent document. Subjects were told that they were free to withdraw from the study at any time, without penalty. The study was conducted according to the Helsinki Declaration, and all procedures used in this study were approved by the Catholic University of Murcia's Review Board before the initiation of the study.

Procedures

Subjects performed 2 similar sets of tests before and after the 6-week training period. The first set of tests was conducted on 2 nonconsecutive days during the week before the beginning of the training program. The second set of tests was conducted under the same conditions during the week after completion of the training program. Both sets of tests were performed using the same procedures, and with the same technician, who was blind to the training-group affiliation. All subjects were familiarized with the testing procedures 1 week before. Before each set of tests, the subjects performed a standard warm-up that included 8 minutes of jogging, followed by 10 minutes of dynamic stretching exercises. The different tests were conducted with a rest interval of 20–30 minutes in between. All tests were performed at the same location and under similar environmental conditions as in the training sessions. The test-retest measurements performed in our laboratory confirmed good reproducibility of all measurements: intraclass correlation coefficient (ICC) >0.85, and typical error of measurement values ranging from 2.6 to 8.3%.

Vertical Jump Tests

Lower limb explosive power was assessed by a CMJ with hands on the hips. Subjects began each jump in an erect standing position, and moved into a semisquat position before jumping as high as possible. The subjects performed 3 trials for each jumping form with passive recovery of 60 seconds between jumps, and the best result was recorded. The test (CMJ) was performed on a Kistler BA9281 platform (Kistler Instrument, Winterthur, Switzerland) installed at ground level. Ground reaction forces (GRFs) were recorded with a sampling frequency of 1,000 Hz. All data were collected on a personal computer for further processing and analysis. Vertical acceleration (from the GRF) was evaluated to obtain the vertical velocity and displacement of the center of mass, using the double integration method (7). The height of the jump was obtained from the velocity value at the moment of take-off using the following equation: $H = \frac{v^2}{2g}$; where v is the take-off velocity and g the gravitational acceleration. The peak power was obtained from the product of force and vertical velocity.

Maximal Strength Test

One repetition maximum, for the bench press and the high pull, was measured on a separate day. The tests were performed using a Multipower machine (Technogym SpA) in which the barbell was attached to both ends, with linear bearings on 2 vertical bars allowing only vertical movement. Warm-up consisted of 10 repetitions set at a load of 50% of the perceived maximum weight. Thereafter, 4 to 5 separate single attempts were performed, starting from a weight of approximately 80% of the maximum perceived. The last acceptable extension with the highest possible load was determined as 1RM. The rest period between attempts was always 3 minutes.

Peak Power Test

The peak power measurement, for the bench press and the high pull, was measured on a separate day. On the Multipower machine (Technogym SpA), a linear position transducer (LPT) (Chronojump Bosco System, Barcelona, Spain) attached to the barbell and interfaced with a computer allowed the recording of bar position with an accuracy of 0.001 seconds. The system was calibrated before the testing session, and bar velocity was subsequently calculated. The LPT produced a voltage signal representative of the degree at which the LPT was extended allowing for displacement-time data to be calculated (8). Instantaneous vertical velocity (v) of the bar throughout the movement was calculated using the displacement (x) and time (t) data at each sample, ($v = \frac{\Delta x}{\Delta t}$). Acceleration of the system (a) was calculated using a second-order derivative of the displacement data ($a = \frac{\Delta x}{\Delta t}$). Force ($F$) produced during the lift was determined by adding the acceleration of the system and acceleration resulting from gravity ($g = 9.81\ \mathrm{m \cdot s^{-2}}$) and then multiplying the total acceleration to the mass of the system (SM), $F = SM \cdot (a + g)$ (8). After these calculations, power (P) was determined by multiplying force and velocity at each time point ($P = F \cdot v$) (12). The production of mechanical power of each repetition in bench press or high pull during the concentric phase of the movement was calculated. The peak power of the best repetition in each situation was used in the analysis. Thirty minutes after testing for the maximum strength, the subjects were asked to perform 4 sets of 3 repetitions of bench press using resistances of 30, 40, 60, and 70% of 1RM; or 5 sets of 3 repetitions of high pull using resistances of 30, 40, 60, 70, and 80% of 1RM, with 3 minutes passive rest between sets. The subjects were spotted by an experienced lifter to ensure that maximum velocity was achieved safely, and the subject was confident under the weight. Loud verbal encouragement was given throughout. The eccentric phase of the lift was performed over 2 seconds and was timed by a digital metronome, whereas the concentric phase was performed at maximum velocity.

Maximal Aerobic Speed Test

An incremental treadmill exercise test was used to assess MAS (3). After 5-minute warm-up running at 6 $\mathrm{km \cdot h^{-1}}$ at 1% inclination, the treadmill velocity was increased to 8 $\mathrm{km \cdot h^{-1}}$, and was increased to 1 $\mathrm{km \cdot h^{-1}}$ at each 2-minute stage until voluntary exhaustion; subjects were encouraged to continue to exercise as long as possible. The test was finished when the subject was not able to maintain the imposed running speed. The speed at the last completed stage was kept as the MAS ($\mathrm{km \cdot h^{-1}}$).

Wingate Test

Wingate test was used to diagnose an anaerobic power output of the tested groups. The test was conducted on a cycle ergometer Technogym Bike Med (Technogym SpA). This test consisted of a 30-second maximal sprint against a constant braking resistance dependent on the subjects' body mass (0.075 $\mathrm{kg \cdot kg^{-1}}$ body mass) according to the optimization tables of Bar-Or (1). The test began from a rolling start, at 60 rpm against minimal resistance. When a constant pedal rate of 60 rpm was achieved, a countdown of "3-2-1-go!" was given by the investigator. Then, subjects were instructed to pedal as fast as they could during 30 seconds. During the test, they were strongly and vigorously exhorted to sprint maximally throughout the 30 seconds. The power produced was calculated as the highest value chosen for maximal power. The average of all measured power values during the Wingate test was taken as mean power. The values were expressed in absolute values of peak power expressed in Watt (W) or relative peak power per body weight expressed in Watt per body weight ($\mathrm{W \cdot kg^{-1}}$).

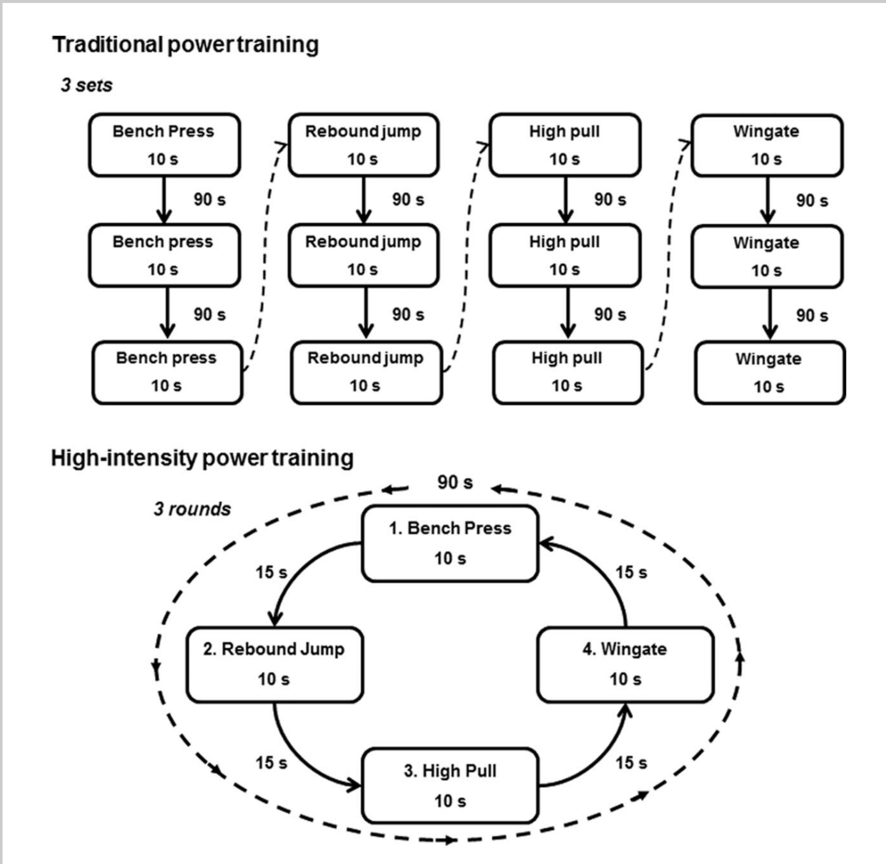

Figure 1. Training and set configuration.

Training Programs

Subjects performed power training 3 times per week for 6 weeks with at least 1 day of rest between each training day. The training volume increased from 3 to 5 sets during the whole training period. One set was added every 2 weeks. Subjects completed each exercise using the optimal load in bench press (i.e., ~30% of 1RM) and high pull (i.e., ~60% of 1RM) exercises, without load in rebound jumps and against a constant braking resistance dependent on the subjects' body mass (0.075 $kg \cdot kg^{-1}$ body mass) on the cycle ergometer. In every session, the subjects performed each exercise to maximum velocity. In bench press and high pull the load was adjusted in the third week. These exercises were chosen to emphasize major muscle groups, using multijoint exercises. All training sessions were supervised by a Certified Strength and Conditioning Specialist (CSCS) to ensure that volitional fatigue was achieved safely, and the control of the rest was strict.

The content of each program was the same concerning exercise intensity and volume during a given training phase. Subjects in both experimental groups completed 10 seconds of each exercise which allowed to perform a range between 5 and 10 repetitions depending on the exercise, as has been prescribed before (24). Although in this study the repetitions number has not been recorded during the training period, in a previous study of our laboratory (34), no differences were observed in the number of repetitions performed during a session of TPT and HIPT using the same protocol. Traditional power training subjects were required to repeat each exercise 3 to 5 sets before moving on to the next exercise following the next order: bench press, rebound jumps, high pull, and 10-second Wingate. The rest between exercises was 90 seconds (Figure 1). This period after each set allows for recovery, through oxidative phosphorylation, of the ATP and phosphocreatine expended during each set (30). During the HIPT program, the subjects executed the training using a short circuit. Circuit training consisted of a series of exercises performed one after another. Subjects were required to perform 3 to 5 rounds using the next sequence: bench press, rebound jumps, high pull, and 10-second Wingate. The rest between exercises was 15 seconds which allowed enough time to move safely between exercises (Figure 1). The total training time in the TPT group ranged between 18 minutes (3 series) and 30 minutes (5 series), whereas in the HIPT group, the total training time ranged between ~5 minutes (3 series) and ~8 minutes (5 series).

Statistical Analyses

Statistical analyses of data were performed using SPSS 21.0 for iOX (IBM Corp. Armonk, NY, USA). Subjects' physical characteristics are reported as mean ± *SD*. The normal distribution and homogeneity parameters were checked using Shapiro-Wilk and Levene tests, respectively. A 2-way repeated measures (RMs) analysis of variance (ANOVA) with TIME (pre- to post-test) and GROUP (TPT vs. HIPT vs. CG) as factors were performed to analyze the training-related effects. If there were statistically significant differences ($p \leq 0.05$) for the time factor, pairwise comparisons were performed to assess pre- to post-test differences for each individual group. Then, 1-way ANOVA with Tukey's pairwise post hoc comparisons were performed on the pre- to post-trial change scores (normalized values relative to the pretest). The ICCs and typical error of measurement were used to determine the test-retest reliability of the dependent variables. Significance was accepted when $p \leq 0.05$. Power (1-β) was determined for all variables, and effect sizes were reported as partial eta-square (η_p^2) and Cohen's *d*. The following criteria for effect sizes were used: <0.35 = trivial, 0.35–0.8 = small, 0.8–1.5 = moderate, and >1.5 = large (35).

Results

The study was based on 29 subjects who were randomly allocated to HIPT ($n = 10$), TPT ($n = 10$), and CG ($n = 9$). Pretraining characteristics of the subjects in each training group are presented in Table 1. No significant differences in any of these characteristics were found between HIPT, TPT, and CG at the beginning of exercise training. No significant differences were observed in training compliance between HIPT and TPT (95.6 ± 2.3 vs. 94.6 ± 2.9%, respectively).

Vertical Jump Test

The height and peak power of the CMJ are presented in Table 2. Two-way ANOVA with RMs revealed a significant time effect for the HIPT ($F = 11.915$; $p = 0.002$; $\eta_p^2 = 0.33$; 1-β = 0.91) and TPT ($F = 9.820$; $p = 0.005$; $\eta_p^2 = 0.29$; 1-β = 0.85) groups, but not CG ($F = 0.944$; $p = 0.341$; $\eta_p^2 = 0.04$; 1-β = 0.15), for CMJ height. Significant differences were found in peak power of the CMJ between pre- to post-test in the TPT group ($F = 7.357$; $p = 0.012$; $\eta_p^2 = 0.24$; 1-β = 0.74). There were no changes in HIPT group ($F = 1.494$; $p = 0.233$; $\eta_p^2 = 0.06$; 1-β = 0.22) and CG ($F = 0.001$; $p = 0.990$; $\eta_p^2 = 0.01$; 1-β = 0.05). One-way ANOVA showed that for CMJ height, the increase in HIPT was greater than in CG ($p = 0.012$; $d = 1.30$). There were no significant differences in the change in height between HIPT and TPT ($p = 0.734$; $d = 0.45$), and TPT and CG ($p = 0.062$; $d = 1.35$).

Maximal Strength Test

One repetition maximum for the bench press and the high pull are presented in Table 2. In bench press, 2-way ANOVA with RMs revealed a significant effect of time for the TPT group ($F = 5.079$; $p = 0.034$; $\eta_p^2 = 0.17$; 1-β = 0.58). There were no changes in HIPT group ($F = 2.389$; $p = 0.135$; $\eta_p^2 = 0.09$; 1-β = 0.32) and CG ($F = 0.027$; $p = 0.871$; $\eta_p^2 = 0.01$; 1-β = 0.05). In high pull, 2-way ANOVA with RMs revealed a significant effect of time for the HIPT ($F = 15.055$; $p = 0.001$; $\eta_p^2 = 0.37$; 1-β = 0.96) and TPT ($F = 48.555$; $p = 0.0001$; $\eta_p^2 = 0.65$; 1-β = 1.00) groups, but not CG ($F = 0.016$; $p = 0.900$; $\eta_p^2 = 0.01$; 1-β = 0.05). One-way ANOVA showed that for high pull exercise, the increase in HIPT was greater than in CG ($p = 0.033$; $d = 1.60$), and the increase in TPT was greater than in CG ($p = 0.0001$; $d = 2.53$).

Peak Power Test

In relation to gains in the peak power (Table 2), there were no differences between pre- to post-test in any groups in bench press exercise (HIPT: $F = 3.686$; $p = 0.066$; $\eta_p^2 = 0.13$; 1-β = 0.46; TPT: $F = 1.581$; $p = 0.220$; $\eta_p^2 = 0.06$; 1-β = 0.23; CG: $F = 1.806$; $p = 0.191$; $\eta_p^2 = 0.07$; 1-β = 0.25). In relation to gains in the peak power in high pull, there were differences between pre- to post-test in training groups (HIPT: $F = 30.924$; $p = 0.0001$; $\eta_p^2 = 0.54$; 1-β = 1.00; TPT: $F = 56.466$; $p = 0.0001$; $\eta_p^2 = 0.69$; 1-β = 1.00), but not CG ($F = 2.441$; $p = 0.130$; $\eta_p^2 = 0.09$; 1-β = 0.33). One-way ANOVA showed that for high pull exercise, the increase in HIPT was greater than in CG ($p = 0.0001$; $d = 2.24$), and the increase in TPT was greater than in CG ($p = 0.0001$; $d = 3.93$).

Maximal Aerobic Speed Test

Two-way ANOVA with RMs revealed a significant effect of time for changes in MAS ($F = 13.639$; $p = 0.001$; $\eta_p^2 = 0.34$; 1-β = 0.95) for the HIPT group (Table 2). There were no changes in this variable in TPT ($F = 1.113$; $p = 0.301$; $\eta_p^2 = 0.04$; 1-β = 0.18) or in CG ($F = 0.001$; $p = 1.000$; $\eta_p^2 = 0.01$; 1-β = 0.05). One-way ANOVA showed that the increase in HIPT was greater than in CG ($p = 0.047$; $d = 1.52$).

Wingate Test

Maximum power (Pmax), maximum power relative to body weight (PmaxR), mean power (Pmean), and mean power relative to body weight (PmeanR) are presented in Table 2. In the training groups, an increase was found in the Pmax (HITP: $F = 21.072$; $p = 0.0001$; $\eta_p^2 = 0.44$; 1-β = 0.99; TPT: $F = 11.842$; $p = 0.002$; $\eta_p^2 = 0.31$; 1-β = 0.91), PmaxR (HIPT: $F = 19.649$; $p = 0.0001$; $\eta_p^2 = 0.42$; 1-β = 0.99; TPT: $F = 15.537$; $p = 0.001$; $\eta_p^2 = 0.37$; 1-β = 0.97), Pmean (HIPT: $F = 30.549$; $p = 0.0001$; $\eta_p^2 = 0.53$; 1-β = 1.00; TPT: $F = 27.307$; $p = 0.0001$; $\eta_p^2 = 0.51$; 1-β = 1.00), and PmeanR (HIPT: $F = 28.911$; $p = 0.0001$; $\eta_p^2 = 0.52$; 1-β = 1.00; TPT: $F = 31.816$; $p = 0.0001$; $\eta_p^2 = 0.54$; 1-β = 1.00). One-way ANOVA showed that the

Table 1. Pretraining characteristics of subjects in each training group.*†

	High-intensity power training group ($n = 10$)	Traditional power training group ($n = 10$)	Control group ($n = 9$)
Age (y)	22.7 ± 3.1	23.4 ± 1.9	23.1 ± 2.5
Weight (kg)	75.0 ± 10.3	76.3 ± 6.7	75.2 ± 9.2
Height (cm)	176.2 ± 6.8	177.1 ± 7.6	175.9 ± 8.2
BMI (kg·m^{-2})	24.2 ± 3.1	24.2 ± 2.9	24.4 ± 2.5

*BMI = body mass index.
†Values are given as mean ± *SD*.

TABLE 2. Performance data.*†

	High-intensity power training group ($n = 10$)			Traditional power training group ($n = 10$)			CG ($n = 9$)		
	Pre	Post	%Δ	Pre	Post	%Δ	Pre	Post	%Δ
Countermovement jump									
Height (cm)	32.3 ± 5.8	34.3 ± 4.4‡	6.2§	34.9 ± 3.3	36.7 ± 4.1‡	5.2	30.3 ± 5.8	29.8 ± 5.6	−1.7
Peak power (W)	3,815.4 ± 766.1	3,926.2 ± 560.8	2.9	3,791.1 ± 564.1	4,037.2 ± 495.5‡	6.5	3,293.1 ± 542.3	3,294.1 ± 547.9	0.0
One repetition maximum									
Bench press (kg)	72.4 ± 13.8	74.5 ± 12.2	2.9	73.7 ± 10.2	76.7 ± 9.0‡	4.1	71.8 ± 12.7	71.5 ± 9.0	−0.4
High pull (kg)	58.5 ± 11.2	63.4 ± 9.3‡	8.4§	58.3 ± 8.1	67.1 ± 9.1‡	15.1§	55.2 ± 10.2	54.8 ± 9.7	−0.7
Peak power									
Bench press (W)	664.3 ± 108.5	691.4 ± 129.3	4.1	651.4 ± 113.1	674.2 ± 110.0	3.5	680.3 ± 120.5	658.3 ± 120.0	−3.2
High pull (W)	1522.8 ± 208.4	1710.4 ± 273.9‡	12.3§	1444.7 ± 288.1	1698.2 ± 273.9‡	17.5§	1489.2 ± 263.3	1426.8 ± 249.4	−4.8
Maximal aerobic speed									
Speed ($km \cdot h^{-1}$)	17.5 ± 0.8	18.2 ± 0.9‡	4.0§	17.4 ± 1.5	17.6 ± 1.8	1.1	16.8 ± 1.3	16.8 ± 1.7	0.0
Wingate									
Pmax (W)	811.4 ± 121.0	883.7 ± 134.4‡	8.9§	810.9 ± 109.6	865.1 ± 112.9‡	6.7§	720.8 ± 148.5	712.1 ± 150.0	−1.2
PmaxR ($W \cdot kg^{-1}$)	11.0 ± 1.0	11.9 ± 1.0‡	8.2§	10.6 ± 1.0	11.4 ± 0.9‡	7.5§	10.0 ± 1.2	9.9 ± 1.3	−1.0
Pmean (W)	662.9 ± 85.9	706.9 ± 96.4‡	6.6§	637.9 ± 62.2	679.5 ± 66.8‡	6.5§	585.9 ± 97.5	581.4 ± 95.6	−0.8
PmeanR ($W \cdot kg^{-1}$)	9.0 ± 0.6	9.6 ± 0.4‡	6.7§	8.4 ± 0.3	9.0 ± 0.4‡	7.1§	8.1 ± 0.7	8.1 ± 0.7	0.0

*Δ = change; Pmax = maximum power; PmaxR = maximum power relative to body weight; Pmean = mean power; PmeanR = mean power relative to body weight; CG = control group.
†Values are given as mean ± *SD*.
‡Significant difference from pre- to post-training ($p \leq 0.05$).
§Significantly different from CG ($p \leq 0.05$).

increase in HIPT was greater than in CG in the Pmax (p = 0.004; d = 1.47), PmaxR (p = 0.007; d = 1.28), Pmean (p = 0.001; d = 1.80), and PmeanR (p = 0.001; d = 1.53), and the increase in TPT was greater than CG in the Pmax (p = 0.026; d = 1.27), PmaxR (p = 0.011; d = 1.21), Pmean (p = 0.001; d = 2.75), and PmeanR (p = 0.0001; d = 2.36).

Discussion

This study aimed to determine the short-term effects (6 weeks) of an HIPT program in healthy young active men, and to compare the effects of HIPT to TPT on the aerobic and anaerobic power as well as strength and power performance. Specifically, although we hypothesized that HIPT would lead to significant improvements in cardiovascular fitness, we were particularly interested in determining whether strength and power changes would be comparable with those elicited by TPT. These results clearly demonstrate that HIPT can be as effective as TPT to elicit improvements in vertical jump performance (i.e., height in CMJ), strength and power performance in high pull exercise, and anaerobic capacity (power developed during Wingate test). Interestingly, only HIPT stimulated positive cardiovascular adaptations in young healthy individuals (i.e., MAS). These findings are important because they indicate that HIPT might be a time effective exercise intervention for triggering multiple positive physiological adaptations.

Jumping ability is an especially useful indicator of physical fitness, and CMJ is a well-recognized training exercise used to enhance it (9). As shown in the results of this study, there were significant differences in CMJ performance between pre- and post-test in the experimental groups (HIPT and TPT), whereas CG remained unchanged. These outcomes have been supported by a multitude of investigations demonstrating improved CMJ performance after various lower-body power, plyometric, and weight-training programs (9,24,42). In addition to the existing scientific literature, this is the first study to demonstrate that improvements in CMJ in response to HIPT were similar to those obtained by TPT. This is of special interest for time-efficient training strategies design, as training performed by the HIPT group differed from TPT only in the rest intervals and sequencing between exercises. Although TPT subjects performed the exercises one after the other with 90-second rest intervals between each series, HIPT subjects executed the training in a short circuit. Approximately 15 seconds separated each exercise, which allowed enough time to move safely between exercises. This circuit was performed for 3–5 series, and the total training time ranged between 5 minutes and 8 minutes, as opposed to the 18–30 minutes (4-fold) required for completing the TPT.

It is well documented in the literature (22) that the main mechanism that enhances jump performance from power training is the increase of the ability of individuals to use the neural and elastic characteristics of the stretch-shortening cycle. After power training, the gradient of the power-time curve during the concentric phase may increase as a result of improved acceleration throughout the movement (9). The source of this improved acceleration may be the ability of subjects to generate additional force at the start of the concentric phase, thus increasing the jump performance (4,23). These improvements stem from the neuromuscular adaptations evident with prolonged strength-power training, including increased cross-sectional area of type II fibers, selective motor unit recruitment, improved firing frequency, and synchronization (23,24).

Regarding the results of strength and power, it has been found improvements in both TPT and HIPT groups, in the bench press and high pull exercises. However, the increments found in 1RM and peak power were statistically significant in the high pull exercise (Table 2). However, improvements have been achieved in the bench press (~3–4%), but these improvements were not statistically significant. Nevertheless, the training load in the high pull exercise was much higher (~60% of 1RM) than the load used for the bench press (~30% of 1RM), so we can argue that the training stimulus was more effective to gain muscle strength and power when higher loads were used (6,31).

In addition to the changes in muscular power and strength, these results also revealed an improvement in mechanical power during the Wingate test. These results agree with the study conducted by Burgomaster et al. (5) who found increases in maximum power after only 2 weeks of sprint interval training (SIT) based on Wingate test, or the study conducted by Hazell et al. (20) who found significant increases in anaerobic performance in Wingate test with only 3 weeks of SIT with bouts of 10 seconds of maximal sprint cycling, similar to our study. These improvements suggest an impact of this training model on anaerobic metabolism and could be attributed to a higher content of muscular phosphocreatine (37), increased activity of the anaerobic enzymes (33), or changes in the motor unit recruitment pattern (10).

Although the main aims of the study were to compare changes in strength and power elicited by HIPT to those achieved with TPT, we also hypothesized that HIPT would promote substantial improvements in cardiovascular fitness. Such improvements were clearly seen, despite the shorter training sessions performed by the subjects in HIPT compared with TPT associated with the need for a high-power production in HIPT. These findings show that aerobic fitness level could be improved through HIPT, which is in agreement with previous findings which showed an increase of 13.6% in $\dot{V}O_2$ max after 10 weeks of similar HIPT training program (40), and are also in line with most of studies that analyzed the effects of SIT, in which improvements have been shown not only in anaerobic metabolism, but also in aerobic power (2,20) for moderately trained subjects. This improvement in MAS could be attributed to an increase in $\dot{V}O_2$ max provoked by peripheral adaptations (26), like enhancement of oxidative muscle capacity, may be due to improvements in mitochondrial density or volume and local enzymatic adaptations as have been seeing after SIT programs (5,16). Therefore, despite the anaerobic nature of the HIPT, the aerobic fitness level performance has shown to increase with only 18 sessions of 3–5 minutes. This suggests that HIPT, like SIT and other forms of HIIT (39), might be considered as a time-efficient training modality of training in comparison with continuous endurance training to improve MAS in active young adults.

In conclusion, we found that both TPT and HIPT are useful training methods to increase power output and strength in moderately resistance-trained young men. Furthermore, HIPT, due to its high intensity and low recovery, involving large cardiovascular demands elicited improvements in MAS, probably due to increments in $\dot{V}O_2$ max similar to those found with different HIIT protocols. These findings are important because they indicate that HIPT training might be a time effective way of triggering multiple positive physiological adaptations in this population. However, our results must be analyzed with caution because power and strength gains might be blunted by the inclusion of shorter interset rest intervals compared with longer resting periods (38). This study shows adaptations of only 6 weeks in moderately trained young men, and although not statistically significant, muscular strength and power outcomes were higher in TPT group. Thus, it is possible that significant differences in power and strength outcomes could be reached if training would have been extended in time due to interference phenomenon. Therefore, future studies will be necessary to assess the long-term effects of HIPT and determine when and how long it should be used.

Practical Applications

To our knowledge, no research has been conducted on the aerobic benefits of an HIPT program. The HIPT focuses on HIPT using multiple joint exercises, with little to no focus on traditional aerobic activities. Despite this, our results show that this type of training also provides aerobic benefits, as well as improvements in muscular strength and power. It has practical applicability due to the existence of many sports that include strength and power expressions under predominantly aerobic requirements (e.g., soccer, basketball). So, training together both qualities would optimize athletic performance similar to traditional training but more time efficiently.

Acknowledgments

The researchers are indebted to the Fundación San Antonio for financing this research with the Grant reference n: PMAFI/23/14. We are also grateful to Mª del Carmen Agüera, Cristian Marín-Pagán, and all the subjects who participated in this research and made this project possible. The results from this study do not constitute endorsement of the products by the authors or by the NSCA.

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